QPCODE: 25900294

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Value Addition Course (VAC) - MG3VACCHE200 - FORENSIC CHEMISTRY

Instruction to the Examiners:

- This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.
- Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.
- Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.

Part A

Multiple choice questions, Answer all questions , Each question carries 1 mark

1. Naloxone is used as an antidote for poisoning by
   Answer: (c) Opiods
2. Which device is commonly used to detect metallic objects during security checks?
   Answer: (b) Metal detector
3. What is the primary function of a police dog in crime scene investigation?
   Answer: C. Track scent trails
4. Human hair is considered a reliable biomarker of arsenic exposure because
   Answer: b). It incorporates arsenic from the bloodstream during growth
5. What is the principle involved in detecting the presence metals using metal detector devices?
   Answer: (c) Generation of electromagnetic field
6. The typical yellow color of urine is primarily due to which of the following?
   Answer: (c) Urochrome
7. Which of the following is an example of a technical document?
   Answer:A. User guide for software
8. Cadmium accumulation in fish particularly affects:
   Answer: (c) Kidneys and liver
9. Which explosive is used for blasting in quarries?
   Answer: (b) Gelatin Stick

Part B

Short Answer Questions- Answer any 5 questions, Each question carries 4 mark

10. **State four signs and symptoms of cyanide poisoning.**
    Answer: Headache, Dizziness or light-headedness, Shortness of breath / rapid breathing, Confusion, or loss of consciousness, Nausea and vomiting, Seizures or convulsions, Low blood pressure and irregular heartbeat.
    (Any 4 points - Give 4 marks)
11. **Differentiate between toxic and corrosive chemicals?**
    Answer: Toxic chemicals are those harmful to living organisms or tissues by intake or absorption into the body by means of consumption, inhaling. These may cause damage to the internal organs and other functional systems like nervous system or digestive system. Eg. lead, mercury, arsenic, chloroform etc.
    Corrosive chemicals are those chemicals which causes destruction on tissues which comes under direct contact. It may causes burn, itching etc. at the point of contact and the effect may be rapid. Eg. sulfuric acid, hydrochloric acid, sodium hydroxide etc. Corrosive chemicals can destroy not only tissues but materials like plastics, metals etc. as well.
    (Any other relevant explanation in 2-3 sentences for each with examples can be given full marks)
12. **Explain the main security features in Indian coins**
    Answer: All students may write alloy composition or any 4 security features .
    (Give 4 marks)
13. **How can DNA fingerprinting be employed in transplantations?**
    (Beyond the scope of the syllabus)

14. **List four postmortem findings in strychnine poisoning.**
Answer: Stiff or rigid muscles / body muscles tightly contracted), Body arched backward ,Bluish lips and nails /cyanosis due to lack of oxygen), Bitter substance in the stomach / strychnine, Frothing at the mouth, Dilated pupils / wide-open eyes, Signs of muscle spasms /contracted jaw, clenched fists. (Any 4 points - Give 4 marks)
15. **Define an incendiary bomb and state its primary purpose.**
Answer: An incendiary bomb is a "firebomb", a weapon designed to start fires and inflict damage through heat and flames rather than a heavy blast. These devices use a slow-burning chemical reaction to release energy over a longer period. Eg. Napalm bomb, white phosphorous etc. Its primary purpose is to attack and produce catastrophic effect on both humans and infrastructure. (Explanation in 3 – 4 sentences)
16. **What are the physical side effects of long-term anabolic steroid use? Mention at least four.**
Answer: (Beyond the scope of the syllabus)

Part C

Short Essay Questions- Answer any 3 questions, Each question carries 7 marks

17. **Discuss the classification of poisons according to their origin with examples.**
Answer: Poisons are substances that can cause harm, illness, or death when introduced into the body. They can be classified based on their **origin**, which helps in understanding their source, mode of action, and treatment. Plant Poisons, Animal Poisons, Mineral or Inorganic Poisons, Chemical (Organic or Synthetic) Poisons. (Any Resonable explanation in 4 parts – 4 marks, examples – 3 marks)
18. **Describe with a diagram the main parts of a bullet and their composition**
The major components of bullets are
(a)Case:- The container that holds all the other ammunition components together. It's usually made of brass, steel, copper, paper, or plastic.
(b)Gunpowder:- A chemical mixture that burns rapidly and converts to an expanding gas when ignited.
(c)Primer:- An explosive chemical compound that ignites the gunpowder when struck by a firing pin. Primer may be placed either in the rim of the case (rimfire) or in the center of the base of the case (centerfire).
(d)Projectile: The bullet is the actual projectile that leaves the end of the barrel and moves towards the target.
(Diagram – 3 marks. Reasonable explanation of at least 3 parts – 4 marks)
19. **Describe the principle and different steps of using a diamond tester for diamond authentication.**
Ans: Diamond Tester
A diamond tester is a handheld device used to verify whether a gemstone is a real diamond or a simulant (like cubic zirconia or moissanite). It works by measuring thermal conductivity (and sometimes electrical conductivity).

Steps to Use a Diamond Tester
1. Turn on the Tester
2. Clean the Gemstone
3. Touch the Probe Tip to the Gemstone
4. Read the Result: If the stone is a real diamond, the device will beep and show a high reading (usually in the red zone). If the stone is not a diamond, it will show a low reading or stay in the green zone. (Principle-2 marks, Steps involved- 5 marks)

20. **Explain the application of ultraviolet rays in forensic document examination with at least four examples.**
Answer:
- Detection of altered inks: Different inks fluoresce differently under UV light, helping to identify additions or modifications.
- Identification of erasures: Erased or rubbed areas often appear as dull or discolored patches under UV light due to damage to paper fibers.
- Distinguishing between original and added entries: UV light can reveal differences in ink used for original text and later additions or corrections.
- Detection of invisible ink: Some forgers use invisible ink that can be detected only under UV light.
- Verification of security features: UV light reveals hidden security features like UV-fluorescent fibers, threads, and seals in currency, cheques, and official documents.
- Detection of washed documents: Chemical washing used to remove original writing leaves traces visible under UV light.

- Authentication of passports and ID cards: Many identity documents contain UV-reactive features to verify authenticity.
- Examination of overwriting or additions: UV light helps identify overwritten texts or figures that are otherwise visually similar.

(Any 4 applications and explanation- Give 7 marks)

21. Explain different types of fibre analysis conducted in forensic labs

Ans: Microscopic Analysis, Spectroscopy, Chromatography, Thermal Analysis (DSC, TG, Wet Chemical Analysis, Burning Tests. Explanantion of any 2 Types of analysis - Give 7 marks

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